

DIFFERENCE-AWARE RETRIEVAL POLICIES FOR IMITATION LEARNING

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ABSTRACT

Parametric imitation learning trained via behavior cloning exhibits degraded generalization on out-of-distribution states, a failure mode attributable to the compounding of single-step prediction errors over the course of closed-loop deployment. We establish that retaining the training distribution as an explicit inference-time resource, through a semi-parametric retrieval-based formulation of imitation learning, mitigates this degradation. We introduce **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, a semi-parametric retrieval-based estimator that reparameterizes the imitation objective in terms of local neighborhood structure over the demonstration manifold rather than a direct global state-to-action map. Rather than fitting a single global policy function, DARP fits a neighbor-conditioned predictor whose inputs are, for each of the k nearest expert states retrieved for a query, the neighbor state, its associated expert action, and the difference vector between the neighbor state and the query state; the resulting per-neighbor action candidates are combined by a permutation-invariant aggregation operator. DARP introduces no assumptions beyond those already required by standard behavior cloning: it requires neither additional data collection, nor online expert feedback, nor task-specific structural priors. Across continuous control and robotic manipulation domains, and across both low-dimensional and high-dimensional visual state representations, DARP yields consistent improvements of 15–46% over standard behavior cloning. Code and demonstrations will be released upon acceptance.

1 INTRODUCTION

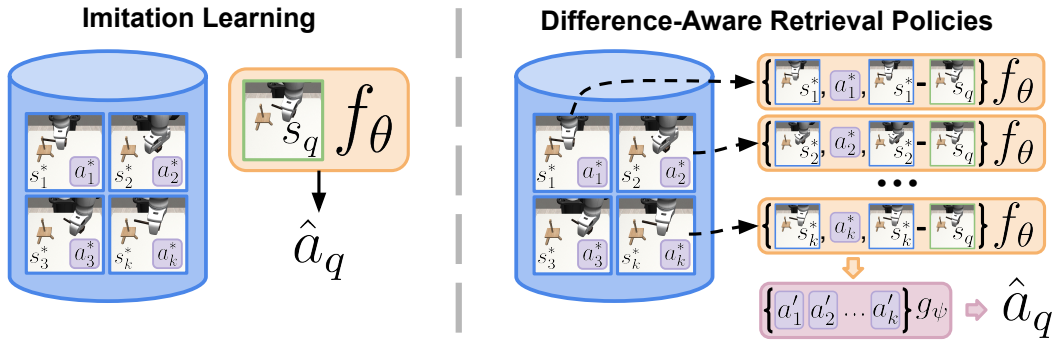


Figure 1: **Overview of DARP:** In contrast to standard BC (left), DARP (right) adopts a retrieval-based reparameterization built upon difference vectors between the query state and its retrieved neighbors. Standard behavior cloning consumes the expert state-action dataset solely during training and discards it at inference time; DARP instead retains the dataset and queries it at inference time, retrieving the local neighborhood of expert state-action pairs surrounding each query point s_q .

Imitation learning via behavior cloning (BC) (Pomerleau, 1991) casts policy acquisition as supervised regression from expert demonstrations, and has enabled robots to acquire complex, dexterous behaviors (Zhao et al., 2023; Chi et al., 2024; Black et al., 2024; Chung et al., 2014). Despite the simplicity of this formulation, BC policies are empirically brittle, particularly on long-horizon tasks (Ross

et al., 2011). The underlying mechanism is *covariate shift*: because the policy is trained under the expert’s state distribution but evaluated under its own induced distribution, per-step prediction errors accumulate over a rollout and drive the agent into states that are underrepresented in the demonstration data (Spencer et al., 2021; Ross et al., 2011). Within these out-of-distribution regions the learned map is poorly constrained, and the resulting policy exhibits high-variance, unreliable behavior that frequently terminates in task failure.

This failure mode is well documented, and numerous methods have been proposed to bound the compounding error (Ross et al., 2011; Venkatraman et al., 2015; Ke et al., 2024b; Levine et al., 2020). Such methods, however, relax the standard BC assumptions, requiring access to a simulator, an interactive expert, large volumes of sub-optimal trajectories, or strong task-specific structure. Our objective is to remain strictly within the pure BC regime: to learn from expert state–action pairs alone, with no supplementary supervision or environment interaction. We therefore pose the following question: *can the variance of a BC policy be reduced using only the original demonstration dataset?*

From a statistical perspective, BC minimizes the supervised risk evaluated at expert states alone. This controls the estimator bias on the training distribution but leaves its variance unconstrained: in low-density regions of the state space, which are precisely the regions encountered under closed-loop rollout, the learned map can oscillate arbitrarily between training points. A principled remedy is to impose a *smoothness* constraint, so that states that are close in the input space are mapped to similar predicted actions (Kobayashi, 2022; Asadi et al., 2018; Ke et al., 2024a; Chen et al., 2024). Such a constraint suppresses spurious high-frequency fluctuations and improves rollout stability; we survey the space of smoothness-inducing mechanisms in Section 4.

While each such mechanism is effective under specific conditions, each also carries a characteristic limitation: data augmentation provides no formal consistency guarantee, global smoothness priors tend to blur behaviorally distinct regions of the state space, temporal penalties regularize only along the trajectory time axis rather than across the state space, and explicit graph regularizers introduce additional smoothness hyperparameters that must be tuned per task.

A complementary axis of the literature contrasts global and local estimators. “Global” supervised models (Black et al., 2024; Zhao et al., 2023; Chi et al., 2024) compress the entire demonstration dataset into the parameters of a single function approximator, which is typically brittle under distribution shift. “Local” estimators (Pari et al., 2022; Mansimov & Cho, 2018; Salzberg & Aha, 1994) instead condition each prediction on the local structure of the dataset, retrieving a neighborhood of similar states and producing outputs through non-parametric or semi-parametric operations over the empirical distribution of expert behavior. Locality confers robustness by removing the dependence on a single fitted parametric function, but it introduces its own limitations: its accuracy is contingent on the retrieval distance metric, unweighted averaging over a neighborhood blurs behaviorally distinct actions and cannot represent multimodal action distributions, and conditioning on the neighbors’ absolute states alone constrains generalization.

We introduce **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, which unifies the robustness of local estimators with the stability of regularized global policy learning. At inference time, in place of a single feedforward evaluation of a parametric function on the current query state, DARP (Fig. 1) first retrieves the k nearest neighbors of the query from the training corpus and then emits k action candidates, each conditioned on a tuple comprising a neighbor state, its associated expert action, and the difference vector from that neighbor to the query. A permutation-invariant operator aggregates these candidates into a single action prediction. This design simultaneously grounds predictions in observed data, via the non-parametric retrieval step, and enforces local consistency, via the parametric predictor conditioned on the retrieved set. We show that this construction reduces estimator variance while introducing no assumptions beyond those of standard behavior cloning: no additional data, no online supervision, and no task-specific structure. Spectrally, the neighbor-aggregation operator approximates a Laplacian filter on the k -nearest-neighbor (k -NN) graph of expert states, yielding a parameter-free smoother that adapts to the local density and geometry of the demonstration manifold.

We support these claims with both theoretical and empirical analysis, establishing that DARP substantially improves performance, by reducing variance and increasing robustness to distribution shift, while operating under the same assumptions as behavior cloning. Our analysis makes the connection to Laplacian regularization precise, showing that DARP implicitly applies a fixed low-

pass spectral filter that attenuates high-frequency variance. Empirically, DARP attains 15–46% gains over standard behavior cloning across continuous control (MuJoCo), robotic manipulation (Robosuite, Robocasa), and high-dimensional visual imitation (Robosuite with image observations). We further show that DARP constitutes a general, scalable architecture that extends to image-based domains and admits expressive policy classes, including transformers and Gaussian mixture models. Finally, we conduct a systematic ablation isolating the contribution of each representational and architectural choice, yielding transferable design insights for retrieval-based methods in sequential decision-making.

2 DIFFERENCE-AWARE RETRIEVAL POLICIES FOR IMITATION LEARNING

In this work we instantiate a class of imitation learning estimators that combines the properties of “global” parametric learning with those of “local” non-parametric learning. We specify an architecture together with a simple training objective that operates under exactly the assumptions of standard behavior cloning while yielding provable and empirical improvements. This section derives the associated theoretical guarantees from first principles. Readers interested primarily in the empirical evaluation may proceed directly to Section 2.5 for a self-contained specification of DARP, followed by Section 3 for the results.

We proceed as follows. Section 2.1 fixes notation and formalizes the problem setting. Section 2.2 introduces a regularized variant of imitation learning that imposes structure derived from the data manifold and establishes improvements in variance, generalization, and stability. Section 2.3 then shows that the effect of this explicit regularizer can be reproduced implicitly by modifying the policy *architecture* in place of the objective. Finally, Section 2.4 specifies our practical algorithm, DARP, which realizes these properties through a semi-parametric, retrieval-augmented architecture applicable to imitation learning with modern neural network and generative-modeling components.

2.1 PRELIMINARIES: BEHAVIOR CLONING FOR IMITATION LEARNING

We adopt the standard imitation learning setting, formalized as a finite-horizon Markov Decision Process (MDP) $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, P_0\}$, where $\mathcal{S}$ denotes the state space, $\mathcal{A}$ the action space, and P_0 the initial-state distribution. A *policy* is a conditional map from states to action distributions, $f_\theta : \mathcal{S} \rightarrow \Delta_{\mathcal{A}}$, parameterized so as to maximize a task-relevant objective; we suppress the subscript θ when referring to the function class f in the abstract. We assume access to a fixed dataset of expert demonstrations $\mathcal{D}^* = \{(s_j^*, a_j^*)\}$ consisting of state-action pairs, and reserve the superscript $*$ in s^* and a^* to denote states and actions drawn from this expert dataset. Behavior cloning (Pomerleau, 1991) recovers a policy f_θ from $\mathcal{D}^*$ by reducing imitation to maximum-likelihood supervised learning, $\arg \max_{\theta} \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\log(f_\theta(a^* | s^*))]$. Although the conditional distribution f_θ may be realized by an arbitrarily expressive generative model (Lipman et al., 2023; Chi et al., 2024), we adopt a Gaussian parameterization in the following exposition for analytical tractability¹.

2.2 WARM-UP: NEIGHBOR MANIFOLD REGULARIZED IMITATION LEARNING

Behavior cloning constrains the policy only through the supervised imitation loss evaluated at expert states drawn from $\mathcal{D}^*$; the objective places no constraint on policy behavior at states that lie off the manifold of expert states. Consequently, the errors that accumulate under closed-loop rollout drive the agent into out-of-distribution regions in which the fitted policy is unconstrained and may take arbitrary values, a phenomenon that is pronounced for overparameterized neural networks (Ross & Bagnell, 2010).

The root cause is that behavior cloning constrains function values only at isolated training states and does not encode the relationship between neighboring states and their corresponding actions, thereby discarding information about the underlying data manifold. To inject this information into the estimator, we augment the objective with a regularization term that explicitly enforces *local consistency*: states that are near one another in the expert dataset should map to similar actions. This yields the neighborhood-regularized loss $\mathcal{L}_{\text{MRIL}}$, in which the standard imitation objective $\mathcal{L}_{\text{BC}}$ is

¹We show that this assumption can be relaxed in Section 2.4

combined with a smoothness penalty $\mathcal{L}_S$ that constrains predictions to respect the local geometry of the dataset rather than relying on pointwise supervision alone.

$$\mathcal{L}_{\text{MRIL}}(f) = \underbrace{\mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)]}_{\text{supervised risk}(\mathcal{L}_{\text{BC}})} + \underbrace{\lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right]}_{\text{smoothness regularizer}(\mathcal{L}_S)}, \quad (1)$$

where $\ell(f(s^*), a^*)$ denotes the supervised imitation loss, $\mathcal{N}_k(s^*)$ indexes the k nearest neighbors of s^* within the expert dataset, and the coefficients $w_i(s^*)$ are normalized kernel weights determined by the state differences, $w_i(s^*) \propto K_\Delta\left(\frac{\|s_i^* - s^*\|}{h}\right)$. As sketched below and established formally in Appendix A.1.1, this penalty is an instance of *manifold regularization*, equivalently Laplacian smoothing, in which the policy incurs a cost for high-frequency variation across the neighborhood of expert states. We prove that this manifold regularizer yields improvements in policy variance, stability, and generalization.

Theorem 1 (Manifold Regularized BC ($\mathcal{L}_{\text{MRIL}}$) improves over vanilla BC ($\mathcal{L}_{\text{BC}}$)). *Let $f^* : \mathcal{S} \rightarrow \mathcal{A}$ be the true, underlying expert policy, assumed to be C^2 -smooth on a compact state space $\mathcal{S}$. Let $f : \mathcal{S} \rightarrow \mathcal{A}$ denote the learned policy estimator. Consider two estimators trained on expert demonstrations:*

1. **Vanilla BC:** a global supervised model minimizing

$$\mathcal{L}_{\text{BC}}(f) = \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)].$$

2. **MRIL:** a neighbor-based estimator minimizing

$$\mathcal{L}_{\text{MRIL}}(f) = \mathcal{L}_{\text{BC}}(f) + \lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right],$$

where $w_i(s^*)$ are the kernel weights defined above and $\lambda > 0$.

Then, under the smoothness assumption on f , the following hold:

- (i) Variance reduction: The Laplacian penalty in MRIL acts as a data-dependent Tikhonov regularizer, yielding smaller estimator variance than vanilla BC.
- (ii) Smoothness guarantee: Minimizers of $\mathcal{L}_{\text{MRIL}}$ satisfy a uniform bound on the local Lipschitz constant of f , whereas vanilla BC admits interpolants with arbitrarily large Lipschitz constants between training states.
- (iii) Policy stability: In a closed-loop rollout, the deviation recursion

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|^2$$

accumulates error linearly for vanilla BC, but sublinearly for MRIL, since the smoothness regularizer enforces $\|f(s^*) - f(s'^*)\| = O(\|s^* - s'^*\|)$ for neighbors s^*, s'^* .

Consequently, MRIL admits strictly stronger generalization and stability guarantees than BC.

Proof sketch. The complete proof is deferred to Appendix A.1.1; we summarize its structure. First, the smoothness regularizer is shown to equal a graph Laplacian penalty on the graph whose k -nearest-neighbor (k -NN) affinity matrix is induced by the kernel weights w_i . Second, in the large-sample limit this graph Laplacian penalty is shown to converge to the weighted Dirichlet energy (Belkin & Niyogi, 2008; Zhou et al., 2003). Minimizing this Dirichlet energy has two consequences: (1) the learned f is locally Lipschitz almost everywhere, which yields smoothness and hence closed-loop stability; and (2) the penalty acts as Tikhonov regularization, reducing estimator variance while keeping the bias term controlled. $\square$

² L_s and L_a are the Lipschitz constants of the environment transition dynamics with respect to state and action, respectively.

The smoothness regularizer does more than penalize pairwise disagreement between neighbors: it drives the learned policy toward smoothness with respect to the underlying data manifold. Specifically, it contracts the local Lipschitz constant of f preferentially along directions in which the data density $p(s)$ is high, so that small perturbations of the state produce small, consistent perturbations of the predicted action. The resulting policy generalizes more reliably on in-distribution (ID) states and extrapolates in a structured manner onto nearby out-of-distribution (OOD) states.

2.3 IMPLICIT MANIFOLD REGULARIZATION VIA IN-CONTEXT ARCHITECTURES

While the MRIL objective amortizes local learning and improves over vanilla BC, it has two drawbacks. First, it introduces a scalar hyperparameter λ that must be tuned to trade off supervised accuracy against smoothness. Second, optimizing a modified regularized objective in place of the standard BC objective can perturb the optimization landscape unfavorably. This motivates the following question: *can the guarantees conferred by MRIL (Eq. 1) be recovered by modifying the policy architecture rather than the objective?*

We answer in the affirmative by introducing a retrieval-based architectural modification that induces an *implicit* manifold regularization effect, denoted iMRIL, while retaining the standard imitation objective. iMRIL recovers the Laplacian smoothing behavior of MRIL under a standard BC objective (illustrated in Fig. 2) and eliminates λ as a training hyperparameter. Section 2.4 then builds on iMRIL to derive the practical algorithm DARP.

iMRIL architecture: The governing principle of iMRIL is to relocate the neighborhood aggregation (averaging) operation from the objective (Eq. 1) into the architecture. Rather than fitting a standard feedforward predictor $f(s)$ against the neighborhood-regularized smoothness objective (Eq. 1), we embed the neighborhood-aggregation structure directly into the parameterization of the action predictor $\hat{f}$, while keeping the objective a standard imitation loss. Concretely, iMRIL fits the parameters of a per-state predictor f_θ such that the aggregated predictor $\hat{f}(s^*) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s^*)} f_\theta(s_i^*)$, formed by averaging f_θ over the nearest-neighbor states $\{s_i^*\}_{i \in \mathcal{N}_k(s^*)}$ from the training set, accurately predicts the corresponding expert action a^* . Under this parameterization, iMRIL optimizes at training time:

$$\arg \min_{\theta} \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} \left[\left\| \underbrace{\left(\frac{1}{k} \sum_{i \in \mathcal{N}_k(s^*)} f_\theta(s_i^*) \right)}_{\hat{f}(s^*)} - a^* \right\|_2 \right] \quad (2)$$

At deployment time, inference for a new query state s_q reduces to retrieving the k -NN of s_q from the training set and applying the neighborhood aggregation $\hat{a} = \hat{f}(s_q) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} f_\theta(s_i^*)$. As Section 2.4 demonstrates, the specific parameterization of f is decisive for the empirical performance of iMRIL, and its refinement is what yields DARP.

In effect, we parameterize the action predictor $\hat{f}$ as an aggregation of the per-neighbor predictions $f(s_i^*)$ over training states and then fit f . Supervising the post-aggregation predictor implicitly precludes f from being arbitrarily non-smooth, conferring the guarantees established in Section 2.2. We prove a direct equivalence between iMRIL and the Laplacian regularization of Section 2.2.

Equivalence between iMRIL and MRIL: We defer the full proof of formal equivalence between MRIL and iMRIL to Appendix Section A.1, and state the main result together with a proof sketch here.

Theorem 2 (iMRIL is parameter-free Laplacian regularization for BC (MRIL)). *Consider the symmetric normalized k -NN graph Laplacian L (defined in Section 2.2), with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$.*

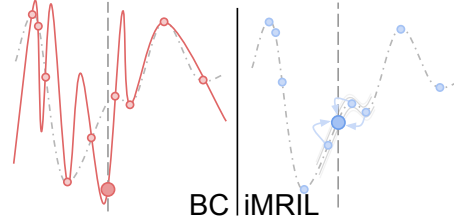


Figure 2: **iMRIL implicitly realizes Laplacian smoothing, reducing variance and enforcing local consistency**, whereas standard BC, lacking any smoothness constraint, admits arbitrarily jagged function approximations.

The minimizers of the explicit MRIL objective (Section 2.2) and the implicit iMRIL objective (Section 2.3) have the following closed form expansions

$$f_{\text{MRIL}} = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j \quad \hat{f}_{\text{iMRIL}} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j$$

iMRIL’s neighbor aggregation step applies the fixed spectral filter $\phi_{\text{iMRIL}}(\mu) = 1 - \mu$ to the graph Laplacian L , preserving low-frequency modes and suppressing high-frequency modes. The congruence between $\hat{f}_{\text{iMRIL}}$ and f_{MRIL} shows that iMRIL is equivalent to a built-in form of Laplacian smoothing (MRIL) with effective $\lambda \approx 1$ in normalized units. Unlike explicit regularization, this implicit filter requires no additional hyperparameter tuning.

Proof sketch. We defer full details to Appendix Section A.1.2. The explicit regularizer admits a spectral solution by diagonalizing the k -NN Laplacian, yielding a filter of the form $(1 + \lambda \mu)^{-1}$ on each eigenmode. The implicit objective can be expressed as neighbor aggregation $\hat{f} = S f$ with $S = D^{-1} A$, the random-walk matrix, which has the same eigenvectors and applies the fixed filter $1 - \mu$. Intuitively, both act as low-pass filters on the graph: modes with small eigenvalues (smooth variation across the data manifold) are largely preserved, while modes with large eigenvalues (rapid, high-variance fluctuations between neighbors) are strongly damped. Thus iMRIL implicitly performs Laplacian smoothing, reducing variance and enforcing local consistency without needing to tune λ . $\square$

The implicit Laplacian smoothing interpretation does not obviate policy learning; rather, it restricts the function class that is representable after aggregation. The neighbor-conditioned network f_θ models how expert actions vary under local state perturbations, emitting a locally adapted action for each neighbor, while the aggregation operator reduces variance by smoothing these per-neighbor proposals across the neighborhood. Learning thus supplies accuracy by correcting local bias, and aggregation supplies stability by controlling variance.

2.4 DIFFERENCE-AWARE RETRIEVAL POLICIES: A PRACTICAL INSTANTIATION OF iMRIL FOR IMITATION LEARNING

Building on the conceptual framework of iMRIL, we now specify a practical algorithm for large-scale imitation learning. Starting from the objective in Eq. 2, we make deliberate design choices for (1) the parameterization of the predictor and (2) the neighbor-aggregation operator, both of which are decisive for empirical performance.

2.4.1 DIFFERENCE-BASED PARAMETERIZATION OF f_θ

The objective in Eq. 2 does not constrain the parameterization or input representation of f_θ . We observe that the role of the neighborhood aggregation is to model how expert actions vary under local state perturbations, which implies that f_θ should be conditioned on the *difference* between the query state and each neighbor state so as to emit a locally adapted action per neighbor. Accordingly, in **Difference-Aware Retrieval Policies (DARP)** we replace the bare parameterization $f_\theta(s_i^*)$ with one that additionally conditions on the neighbor’s expert action a_i^* and on the difference vector $\Delta s_i = s_i^* - s_q$ between the query and neighbor states. The predictor f_θ thus maps a query state s_q and a neighbor (s_i^*, a_i^*) to an action candidate $a_i' = f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)$.

Let $\mathcal{N}_k(s_q)$ denote the index set of the k nearest neighbors retrieved under a distance function $d(s_q, s_i^*)$.³ DARP generates a prediction for any query state s_q by applying the neighborhood aggregation of Section 2.3:

³We instantiate d as the Euclidean distance in a pre-trained embedding space, though alternative neighborhood functions are equally admissible. Appendix Section A.2.1 discusses the design choices governing neighborhood construction via retrieval in detail.

$$\hat{a}_q = f_{\text{DARP}}(s_q) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} a'_i \quad (3)$$

$$= \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q). \quad (4)$$

At training time, this can be used to define a straightforward imitation learning objective from the expert dataset $\mathcal{D}^*$:

$$\arg \min_{\theta} \mathbb{E}_{(s_q^*, a_q^*) \sim \mathcal{D}^*} [\|f_{\text{DARP}}(s_q^*) - a_q^*\|^2] \quad (5)$$

where the optimization is over the parameters θ of the predictor f_θ and minimizes the squared discrepancy between the aggregated prediction $\hat{a}_q$ and the expert action a_q^* . Because the objective is a standard regression loss, f_θ admits any parameterization; we use standard feedforward or convolutional networks. As Section 3 demonstrates, the difference-based parameterization is critical to performance. At inference time, the executed action is produced by retrieving the k -NN of the query and evaluating the neighborhood aggregation of Eq. 3.

2.4.2 GOING BEYOND LINEAR AGGREGATION

The neighborhood aggregation considered thus far, the unweighted mean $\hat{a}_q = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} a'_i$, is a special case of the broader class of permutation-invariant aggregation operators $g_\psi(\{a'_i\}_{i \in \mathcal{N}_k(s_q)})$. In particular, g_ψ may be instantiated by an expressive set-equivariant architecture such as the Set Transformer (Lee et al., 2019) or DeepSets (Zaheer et al., 2017), which generalizes the prediction model of Eq. 3 to $\hat{a}_q = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)\}_{i \in \mathcal{N}_k(s_q)})$. Beyond increased expressivity, replacing the mean with a parametric aggregation operator g_ψ enables the representation of richer action distributions, such as Gaussian mixture models (Pignat & Calinon, 2019) or diffusion models (Chi et al., 2024), in place of the unimodal Gaussian implicit in the L_2 -regression objective of Eq. 5. Rather than emitting $\hat{a}_q$ directly, DARP can predict the parameters α of an action distribution $p(a_q; \alpha)$, for example the component means, covariances, and mixture weights of a Gaussian mixture model or the score function of a diffusion model. This permits maximum-likelihood training of multimodal action distributions in place of unimodal L_2 regression:

$$\arg \max_{\theta} \mathbb{E}_{(s_q^*, a_q^*) \sim \mathcal{D}^*} [\log p(a_q^*; \alpha_\theta(s_q^*))], \text{ where } \alpha_\theta(s_q^*) = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q^*)\}_{i \in \mathcal{N}_k(s_q^*)}) \quad (6)$$

Inference for a query state s_q proceeds by retrieving its neighbors at test time, constructing $\alpha_\theta(s_q) = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)\}_{i \in \mathcal{N}_k(s_q)})$, and sampling $\hat{a}_q \sim p(\cdot; \alpha_\theta(s_q))$. Detailed training pseudocode is provided in Appendix Section A.3.

2.5 SUMMARY

We have established that enforcing local consistency while fitting a behavior cloning policy is equivalent to manifold regularization (Laplacian smoothing) and provably improves variance, smoothness, and stability relative to standard BC (Theorem 1). We have further shown that, rather than adding an explicit regularization term and its associated hyperparameter λ to the loss, the same smoothing can be induced implicitly by embedding neighborhood aggregation in the policy architecture (Theorem 2). These results yield the following algorithm. At training time, for each expert state-action pair (s_q^*, a_q^*) , DARP retrieves the k nearest neighbor states from $\mathcal{D}^*$, computes the difference vector $\Delta s_i = s_i^* - s_q^*$ from each neighbor to the query, evaluates the predictor f_θ on each neighbor tuple $(s_i^*, a_i^*, \Delta s_i)$ to obtain a candidate action, and aggregates the candidates through a permutation-invariant operator g_ψ to produce the final prediction. The entire estimator is trained under the standard imitation objective alone.

3 EXPERIMENTAL EVALUATION

We evaluate DARP to address three questions: **Q1**: Does DARP consistently outperform standard behavior cloning? **Q2**: Does DARP accommodate more complex state representations and action distributions? **Q3**: What is the marginal contribution of each architectural component to DARP’s performance gains? The evaluation spans multiple domains, covering low-dimensional state representations, high-dimensional image features, and diverse action representations. Concretely, it comprises continuous control (MuJoCo), robotic manipulation (Robosuite), and purpose-built discontinuous environments that stress-test the neighbor-based estimator under adversarial local geometry.

3.1 BASELINE COMPARISONS AND TASK DESCRIPTIONS

MuJoCo Tasks: The MuJoCo (Todorov et al., 2012; Fu et al., 2020) continuous-control tasks require actuating legged morphologies of varying embodiment to achieve forward locomotion on a planar surface, with reward proportional to forward velocity under stability constraints. We evaluate on Hopper (single-legged hopper), Walker (bipedal humanoid), Ant (quadruped), and HalfCheetah (planar biped).

Robosuite Tasks: The Robosuite (Zhu et al., 2020) tasks each require a single robotic arm to manipulate objects. Stack requires placing a smaller cube atop a larger one; Threading requires manipulating a thin needle-like tool and inserting it through a small ring; Square Peg requires grasping a square block with a central hole and seating it onto a square peg.

RoboCasa Tasks: The RoboCasa (Nasiriany et al., 2024) tasks each require a single robotic arm to manipulate objects within a procedurally randomized kitchen scene. Drawer requires closing an open drawer; Door requires closing an open microwave door; Stove requires rotating a knob to extinguish a burner.

Baseline Comparisons: We benchmark DARP against the following baselines and ablations: (1) **R&P** (Sridhar et al., 2025), which executes the action of the single nearest neighbor; (2) **LWR** (Pari et al., 2022), which performs locally weighted regression over the retrieved neighbors; (3) **BC**, standard parametric behavior cloning; (4) **REGENT** (Sridhar et al., 2025), a transformer-based in-context learning method conditioned on retrieved neighbors; and (5) **MRIL**, the explicitly smoothed variant of DARP specified in Section 2.2.

3.2 CAN DARP CONSISTENTLY OUTPERFORM STANDARD BEHAVIOR CLONING? (Q1)

We first evaluate the core hypothesis of DARP on tasks with low-dimensional state representations, a regime in which inter-state distances are well-defined and interpretable. This evaluation spans MuJoCo locomotion together with Robosuite and RoboCasa manipulation, the latter using demonstrations generated by MimicGen (Mandlekar et al., 2023). Throughout this subsection, the aggregation operator g is instantiated as the unweighted mean of the per-neighbor action candidates a' .

Method	Hopper	Ant	Walker	HalfCheetah
R&P (Sridhar et al., 2025)	711.82 $\pm$ 85.63	-305.97 $\pm$ 76.42	419.18 $\pm$ 50.21	-178.64 $\pm$ 29.75
LWR (Pari et al., 2022)	1703.78 $\pm$ 245.95	846.59 $\pm$ 216.06	1484.91 $\pm$ 356.54	1945.82 $\pm$ 567.26
BC	2313.65 $\pm$ 203.75	2376.20 $\pm$ 339.43	2658.40 $\pm$ 274.08	1063.23 $\pm$ 371.08
REGENT (Sridhar et al., 2025)	1819.39 $\pm$ 186.24	-302.10 $\pm$ 146.67	507.01 $\pm$ 76.10	169.85 $\pm$ 63.10
MRIL	2793.63 $\pm$ 156.41	3869.08 $\pm$ 241.00	4370.96 $\pm$ 168.13	701.58 $\pm$ 195.08
DARP	3545.57 $\pm$ 3.54	4383.28 $\pm$ 266.37	4894.01 $\pm$ 75.12	5515.41 $\pm$ 841.33
DARP Set Transformer	2965.86 $\pm$ 103.08	4063.79 $\pm$ 218.80	4752.42 $\pm$ 109.23	3417.85 $\pm$ 764.57

Table 1: **Both DARP and DARP Set Transformer outperform all baselines across every domain.** Return of DARP versus BC and competing baselines on MuJoCo environments under low-dimensional state. Reported values are means over 100 independent trials with 95% confidence intervals. Parametric policies are Multi-Layer Perceptrons; results with diffusion policies are reported in Section 8.

Across every evaluated environment, DARP yields substantial improvements over standard behavior cloning: gains of 15–25 percentage points on the robotic manipulation tasks and large increases in return on the locomotion tasks (Table 1 and Table 2). The purely non-parametric baselines (R&P

and LWR) perform poorly across this suite, and while MRIL nearly always exceeds vanilla BC, the maximum score on every task is attained by the DARP architecture.

We next assess whether the practical instantiation of Section 2.4 scales to higher-dimensional input representations such as images.



Figure 3: The “DrawerClose” RoboCasa task, in which the robotic arm must close the open drawer.



Figure 4: The “Threading” Robosuite task, in which the robotic arm must insert the needle implement through the small hole.

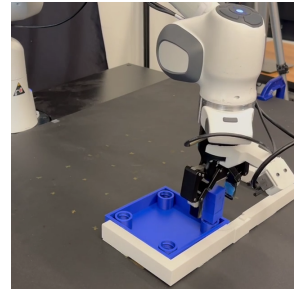


Figure 5: The real-world FurnitureBench square-table assembly task. The robot must grasp a table leg and screw it into a threaded hole at the corner of the tabletop.

3.3 CAN DARP HANDLE MORE COMPLEX STATE REPRESENTATION AND ACTION DISTRIBUTIONS? (Q2)

High-Dimensional Visual Input Representations. To probe the applicability of DARP beyond compact low-dimensional states, we evaluate it on simulated robotic manipulation using R3M image embeddings (Nair et al., 2022), thereby testing whether the neighbor-

Method	Stack	Thrd.	Peg
BC	44	38	17
DARP	75	76	52

Table 3: Success rates (%) on vision-based Robosuite tasks under R3M image embeddings.

Method	RoboCasa			Robosuite			Real
	Drawer	Door	Stove	Stack	Thrd.	Peg	Sq. Table
BC	54	29	28	47	37	46	44
DARP	85	45	43	72	63	62	92

Table 2: DARP versus BC across all three evaluation environments under low-dimensional state features. Entries are success rates in percent (RoboCasa and Robosuite: 100 trials; Real: 50 trials). DARP dominates BC on every task and environment, and more than doubles the BC success rate in the real-world setting. RoboCasa and Robosuite policies are Multi-Layer Perceptrons, whereas the real-world results use a diffusion policy; see Section 8.

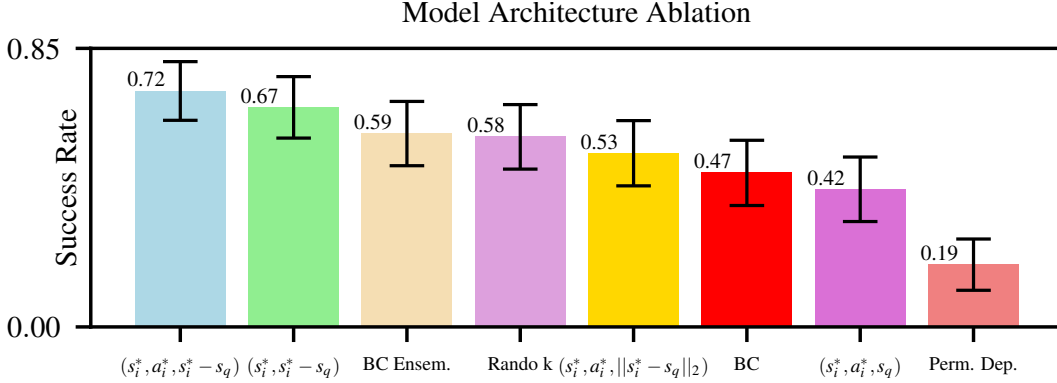


Figure 6: **Difference vectors and permutation invariance account for most of DARP’s performance.** Ablating individual architectural components isolates the difference-vector representation $(s_i^*, a_i^*, s_i^* - s_q)$ as the dominant source of DARP’s gains. Success rate is averaged over 100 trials on the Robosuite Stack environment, with 95% confidence intervals.

based estimator remains effective when states are high-dimensional feature vectors extracted from visual observations (Table 3). DARP not only surpasses standard BC but does so by a larger margin than under low-dimensional state: the mean improvement of $\sim 35\%$ on the visual tasks exceeds the $\sim 22\%$ mean improvement on the corresponding low-dimensional Robosuite tasks. This indicates that DARP adapts to complex, high-dimensional state representations more effectively than standard BC.

Multi-modal Action Distributions. DARP also solves complex multimodal imitation tasks: on the Push-T environment it exceeds behavior cloning by more than 20%. Full details are deferred to Appendix A.2.2.

3.4 HOW DO DIFFERENT ARCHITECTURAL COMPONENTS CONTRIBUTE TO DARP’S PERFORMANCE GAINS? (Q3)

Ablation Study: To quantify the marginal contribution of each component of the DARP architecture, we conduct a systematic ablation over the following variants: (1) full DARP; (2) DARP with the neighbor actions removed from the predictor input; (3) an ensemble of 10 independently trained BC agents; (4) DARP with neighbors sampled uniformly at random rather than by the distance metric; (5) DARP conditioned on the scalar L_2 norm of the difference vector in place of the vector itself; (6) the BC baseline, conditioned on the query state s_q alone; (7) DARP conditioned on the query state in place of the query-to-neighbor difference vector; and (8) DARP with a permutation-dependent (i.e. **not** permutation-invariant) aggregator over the candidates a' . Figure 6 reports the results.

The ablation establishes that the difference-vector representation and permutation invariance are the principal drivers of DARP’s performance, whereas the neighbor actions have a comparatively modest effect. Uniform random neighbor selection degrades performance sharply, confirming that a meaningful distance metric governing retrieval is essential. Permutation invariance of the aggregation

operator g is likewise critical: substituting a permutation-dependent aggregator produces a substantial performance drop.

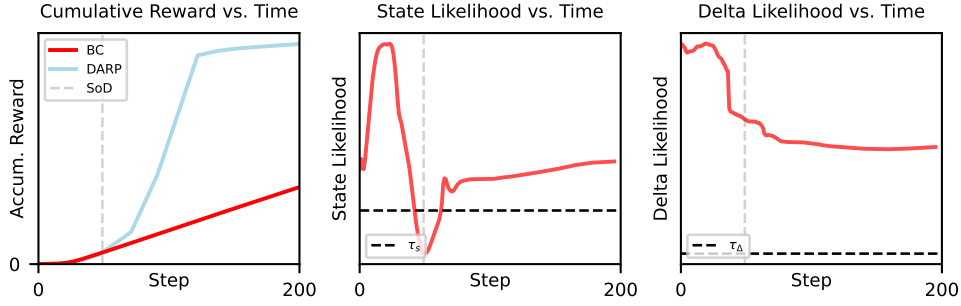


Figure 7: **Cumulative reward of BC and DARP on the Robosuite Stack task: the rollouts are initially identical and diverge at the step where BC fails and DARP succeeds.** The vertical dashed line marks the step of divergence, denoted “SoD”. At the SoD the state likelihood satisfies $< \tau_s$ (out-of-distribution) while the difference-vector likelihood satisfies $> \tau_\Delta$ (in-distribution).

Divergence Analysis: To characterize the mechanism underlying DARP’s advantage over standard BC, we analyze the point of divergence in rollouts where BC fails but DARP succeeds. We define the “step of divergence” (SoD) as the first step at which the two methods accrue significantly different reward. We set the thresholds τ_s and τ_Δ to the 1st percentile of the respective training-set likelihoods; equivalently, 1% of the difference vectors observed at training time have likelihood below τ_Δ . Across all six rollouts spanning two tasks (Robosuite Stack and MuJoCo Hopper), the query state at the SoD has state likelihood $< \tau_s$ yet difference-vector likelihood $> \tau_\Delta$. This dissociation supports the hypothesis that DARP’s gains derive in part from improved prediction on marginally out-of-distribution states, a consequence of reparameterizing the input in terms of difference vectors to neighbors (Figure 7 plots reward drift, the SoD, and the state and difference-vector likelihoods for one task). Appendix A.2.4 reports additional robustness experiments.

4 RELATED WORK

Non-Parametric Imitation Learning Methods: Non-parametric imitation learning algorithms attain strong performance by exploiting local structure in the demonstration data. VINN Pari et al. (2022) applies locally weighted regression to imitation and reports strong results in image embedding spaces, while MiDiGaP von Hartz et al. (2025) models multimodal trajectories via mixtures of Gaussian processes to achieve rapid generalization. SEABO Lyu et al. (2024) employs retrieval for offline RL, constructing a reward function that rewards transitions proximate to their neighbors. FlowRetrieval Lin et al. (2024), STRAP Memmel et al. (2025), and Behavior Retrieval Du et al. (2023) perform non-parametric retrieval and fine-tuning over large unlabeled datasets, enabling generalization through test-time training. DARP is distinguished from these methods by two design choices: it parameterizes each retrieved neighbor as the tuple $(s_i^*, a_i^*, s_i^* - s_q)$, and it fits a semi-parametric policy rather than relying on purely non-parametric aggregation or test-time training. These choices jointly deliver the variance reduction of local estimators and the generalization of parametric policies.

Smoothness-Constrained Policy Learning: A substantial body of recent work imposes explicit smoothness constraints to improve policy stability and robustness. L2C2 Kobayashi (2022) formulates model-free RL under local Lipschitz-continuity constraints, obtaining smoothness and noise robustness without sacrificing expressiveness, and Asadi et al. (2018) develop an analogous Lipschitz-constrained formulation for model-based RL. CCIL Ke et al. (2024a) extends these ideas by generating synthetic corrective labels for imitation learning from a Lipschitz-constrained dynamics model, an approach subsequently scaled to humanoid controllers Chen et al. (2024) to reduce deployment-time jitter. DARP departs from all of these by enforcing smoothness implicitly through an architectural modification while retaining a standard imitation objective.

In-Context Learning Methods: A related line of work approaches non-parametric retrieval through in-context imitation learning. REGENT Sridhar et al. (2025) studies retrieval-augmented generalization by conditioning a causal transformer on retrieved states, actions, and rewards, while DPT Lee et al. (2023) pretrains transformers in a supervised manner to predict actions from a query state

and an in-context dataset, thereby learning an exploration strategy. Further in-context architectures include ICRT Fu et al. (2024), Instant Policy Vosylius & Johns (2025), and KAT Di Palo & Johns (2024). Whereas these methods target rapid adaptation to novel tasks and environments, DARP instead targets higher performance and stability on the standard imitation learning problem.

5 CONCLUSION

We introduced **D**ifference-Aware **R**etrieval **P**olicies for Imitation Learning (**DARP**) (DARP), a nearest-neighbor-based estimator that reparameterizes the imitation objective in terms of the difference vectors between a query state and its nearest neighbors rather than fitting a direct global state-to-action map. We proved that this construction implicitly realizes Laplacian smoothing. The empirical evaluation, spanning continuous control and robotic manipulation, substantiates three hypotheses. First, DARP consistently outperforms standard behavior cloning under low-dimensional state representations. Second, its advantage persists across state representations, action-distribution modeling requirements, and task complexities, with improvements of 15–46% across the evaluated settings. Third, architectural ablations identify the difference-vector representation and the permutation-invariant aggregation operator as the two components essential to the method.

6 REPRODUCIBILITY STATEMENT

Upon acceptance we will release the supplementary source code, the policy and environment configuration files, the MuJoCo experiment data, the scripts that generate expert demonstrations for the Robosuite tasks via MimicGen, and the modality-transformation code. On NVIDIA L40 and L40s GPUs, the released code reproduces the reported results exactly, with the sole exception of results that depend on transformer components (REGENT and the Set Transformer), whose non-deterministic execution may induce minor deviations from the reported numbers.

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A APPENDIX

A.1 LEMMAS AND PROOFS

A.1.1 PROOF OF THEOREM 1

The proof of Theorem 1 builds on the following established result (Chung, 1997; Zhou et al., 2003; Belkin & Niyogi, 2008).

Lemma 1 (Smoothness regularizer as k -NN graph Laplacian penalty). *Let $\{s_1^*, \dots, s_n^*\}$ be the expert states with corresponding predicted actions $f(s_i^*) \in \mathbb{R}^{d_a}$. For each i , let $\mathcal{N}_k(s_i^*)$ denote the indices of the k -nearest neighbors of s_i^* (excluding i). Define asymmetric weights*

$$\tilde{W}_{ij} = \begin{cases} w_j(s_i^*), & \text{if } j \in \mathcal{N}_k(s_i^*), \\ 0, & \text{otherwise,} \end{cases}$$

and construct a symmetric affinity matrix

$$W_{ij} = \frac{1}{2}(\tilde{W}_{ij} + \tilde{W}_{ji}).$$

Let D be the degree matrix with $D_{ii} = \sum_j W_{ij}$, and define the k -NN graph Laplacian $L = D - W$.

Then the smoothness regularizer can be written as the quadratic form

$$\mathcal{L}_S(f) = \frac{1}{n} \sum_{i=1}^n \sum_{j \in \mathcal{N}_k(s_i^*)} w_j(s_i^*) \|f(s_i^*) - f(s_j^*)\|^2 \propto \text{Tr}(F^\top L F),$$

where $F = [f(s_1^*), f(s_2^*), \dots, f(s_n^*)]^\top \in \mathbb{R}^{n \times d_a}$. Equivalently, in the scalar case,

$$\mathcal{L}_S(f) \propto f^\top L f.$$

Corollary 1 (Continuum limit of smoothness regularizer). *Assume states $\{s_i^*\}_{i=1}^n$ are sampled i.i.d. from a smooth density $p(s^*)$ supported on an m -dimensional C^2 manifold $\mathcal{M} \subset \mathbb{R}^d$. Let W be the symmetrized k -NN affinity matrix constructed from a kernel K_Δ with bandwidth h , and let $L = D - W$ be the graph Laplacian.*

If $n \rightarrow \infty$, $h \rightarrow 0$, and $nh^{m+2} \rightarrow \infty$, then the normalized quadratic form converges to the weighted Dirichlet energy:

$$\frac{1}{n^2 h^{m+2}} \text{Tr}(F^\top L F) \rightarrow C_K \int_{\mathcal{M}} \|\nabla_{\mathcal{M}} f(s)\|_2^2 p(s)^2 d\text{vol}(s),$$

where $C_K > 0$ is a constant depending only on the kernel K_Δ .

Theorem 1 (Manifold Regularized BC ($\mathcal{L}_{\text{MRIL}}$) improves over vanilla BC ($\mathcal{L}_{\text{BC}}$)). *Let $f^* : \mathcal{S} \rightarrow \mathcal{A}$ be the true, underlying expert policy, assumed to be C^2 -smooth on a compact state space $\mathcal{S}$. Let $f : \mathcal{S} \rightarrow \mathcal{A}$ denote the learned policy estimator. Consider two estimators trained on expert demonstrations:*

1. **Vanilla BC:** a global supervised model minimizing

$$\mathcal{L}_{\text{BC}}(f) = \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)].$$

2. **MRIL:** a neighbor-based estimator minimizing

$$\mathcal{L}_{\text{MRIL}}(f) = \mathcal{L}_{\text{BC}}(f) + \lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right],$$

where $w_i(s^*)$ are the kernel weights defined above and $\lambda > 0$.

Then, under the smoothness assumption on f , the following hold:

- (i) Variance reduction: The Laplacian penalty in MRIL acts as a data-dependent Tikhonov regularizer, yielding smaller estimator variance than vanilla BC.

(ii) Smoothness guarantee: Minimizers of $\mathcal{L}_{\text{MRIL}}$ satisfy a uniform bound on the local Lipschitz constant of f , whereas vanilla BC admits interpolants with arbitrarily large Lipschitz constants between training states.

(iii) Policy stability: In a closed-loop rollout, the deviation recursion

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|^4$$

accumulates error linearly for vanilla BC, but sublinearly for MRIL, since the smoothness regularizer enforces $\|f(s^*) - f(s'^*)\| = O(\|s^* - s'^*\|)$ for neighbors s^*, s'^* .

Consequently, MRIL admits strictly stronger generalization and stability guarantees than BC.

Proof. We prove each claim in turn.

(i) Variance reduction. The Laplacian penalty in $\mathcal{L}_{\text{MRIL}}$ is

$$\sum_{i,j} W_{ij} \|f(s_i^*) - f(s_j^*)\|^2 = 2f^\top Lf,$$

where $L = D - W$ is the graph Laplacian, and W is the k -NN affinity matrix. By Lemma 1, this equals the empirical Dirichlet energy of f on the k -NN graph. It is well known (Zhou et al., 2003; Belkin & Niyogi, 2008) that such a quadratic penalty is equivalent to Tikhonov regularization with respect to the graph Laplacian norm $\|f\|_L^2 = f^\top Lf$. In statistical learning theory, adding a Tikhonov penalty strictly reduces the variance of the estimator compared to the unregularized solution while keeping the bias term controlled. Thus MRIL enjoys smaller estimator variance than vanilla BC, which uses no such penalty.

(ii) Smoothness guarantee. Consider the continuum limit (Corollary 1): for i.i.d. samples $\{s_i^*\}$ from density p on a smooth manifold $\mathcal{M}$, the normalized penalty converges to

$$\int_{\mathcal{M}} \|\nabla f(s)\|^2 p(s)^2 d\text{vol}(s).$$

This is the weighted Dirichlet energy of f on $\mathcal{M}$. If this integral is finite, f belongs to the Sobolev space $H^1(\mathcal{M}, p^2)$, and in particular f is locally Lipschitz almost everywhere with

$$\|f(s^*) - f(s'^*)\| \leq C\|s^* - s'^*\| \quad \text{for } p\text{-a.e. neighbor pairs } s^*, s'^*.$$

Therefore minimizers of $\mathcal{L}_{\text{MRIL}}$ have uniformly bounded local Lipschitz constants along high-density regions of the state space. By contrast, minimizers of vanilla BC have no such constraint: any oscillatory interpolant that matches the training data exactly yields the same supervised risk, so arbitrarily large Lipschitz constants are possible.

(iii) Policy stability. Let $\Delta_t = \|s_t - s_t^*\|$ denote the deviation at time t . For Lipschitz dynamics T ,

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|.$$

Decompose the action error:

$$\|f(s_t) - f^*(s_t^*)\| \leq \|f(s_t) - f^*(s_t)\| + \|f^*(s_t) - f^*(s_t^*)\|.$$

For vanilla BC, the first term $\|f(s_t) - f^*(s_t)\|$ is only minimized on the empirical distribution $P_{\mathcal{S}}$; off-distribution, it may be $O(1)$ regardless of Δ_t . The second term satisfies $\|f^*(s_t) - f^*(s_t^*)\| = O(\Delta_t)$ by smoothness of f^* . Hence the recursion can take the form

$$\Delta_{t+1} \leq L_s \Delta_t + L_a (O(1) + O(\Delta_t)),$$

which accumulates linearly in t .

For MRIL, the Laplacian penalty enforces

$$\|f(s^*) - f(s'^*)\| \leq C\|s^* - s'^*\| \quad \text{for neighbor pairs } (s^*, s'^*),$$

⁴ L_s and L_a are the Lipschitz constants of the environment transition dynamics with respect to state and action, respectively.

as shown in part (ii). Thus $\|f(s_t) - f^*(s_t)\| = O(h^2)$ (where h is the radius of the kernel bandwidth around the point s_t) by local-linear regression error bounds, and $\|f^*(s_t) - f^*(s_t^*)\| = O(\Delta_t)$ by smoothness of f^* . Combining these,

$$\Delta_{t+1} \leq L_s \Delta_t + L_a (O(h^2) + O(\Delta_t)).$$

Since the constant multiplying Δ_t is strictly smaller under the smoothness constraint, the cumulative error grows strictly slower than in vanilla BC. In particular, error growth is sublinear in the rollout horizon when h is small, whereas it is linear for vanilla BC.

Conclusion. Claims (i)–(iii) establish that MRIL yields lower variance, uniform smoothness control, and sublinear rollout error accumulation compared to vanilla BC, completing the proof. $\square$

Kernel choice. For the smoothness regularizer we adopt the normalized Gaussian kernel

$$w_i(s^*) \propto \exp\left(-\frac{\|s^* - s_i^*\|^2}{2h^2}\right), \quad \sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) = 1,$$

with bandwidth h fixed to the median distance to the k -th nearest neighbor over the dataset. This is the standard choice in manifold regularization (Belkin & Niyogi, 2008; Zhou et al., 2003) and guarantees that the graph Laplacian penalty converges to the Dirichlet energy in the continuum limit. Empirically, this default is stable across tasks, and alternative kernels (e.g. the uniform k -NN or exponential-decay kernels) yield qualitatively equivalent results.

A.1.2 PROOF OF THEOREM 2

We first establish the two lemmas that characterize the spectral form of, respectively, explicit Laplacian regularization and neighbor aggregation.

Lemma 2 (Spectral form of explicit Laplacian regularization). *Let L be the symmetric normalized graph Laplacian with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$. The minimizer of the penalized objective*

$$\mathcal{L}_\lambda(f) = \|f - a^*\|^2 + \lambda f^\top L f$$

has the closed-form expansion

$$f_\lambda = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j.$$

Thus λ directly determines the spectral filter $\phi_\lambda(\mu) = (1 + \lambda \mu)^{-1}$ applied to each Laplacian mode.

Proof. Diagonalize $L = U \Lambda U^\top$ with $U = [u_1, \dots, u_n]$ orthogonal and $\Lambda = \text{diag}(\mu_1, \dots, \mu_n)$. Write $f = U c$, $a^* = U b$ in this basis. The objective becomes

$$\|U c - U b\|^2 + \lambda c^\top \Lambda c = \sum_{j=1}^n (c_j - b_j)^2 + \lambda \mu_j c_j^2.$$

Minimizing each term yields $c_j = \frac{1}{1 + \lambda \mu_j} b_j$. Transforming back gives the stated expansion. $\square$

Lemma 3 (Spectral form of neighbor aggregation). *Let $S = D^{-1}A$ be the random-walk matrix of the k -NN graph, with adjacency A and degree D . For any prediction vector f , the neighbor-averaged prediction is $\hat{f} = S f$. In the Laplacian eigenbasis, this corresponds to the spectral filter*

$$\hat{f} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j,$$

i.e. $\phi_{\text{DARP}}(\mu) = 1 - \mu$.

Proof. By definition, $L = I - D^{-1/2} A D^{-1/2}$ and $S = D^{-1} A = I - L_{\text{rw}}$ where $L_{\text{rw}} = D^{-1} L$ is the random-walk Laplacian. Since L_{rw} and L share the same spectrum up to similarity transform, the eigenbasis $\{u_j\}$ diagonalizes S . Thus for each mode u_j , $S u_j = (1 - \mu_j) u_j$, yielding the claimed spectral filter. $\square$

Theorem 2 (iMRIL is parameter-free Laplacian regularization for BC (MRIL)). Consider the symmetric normalized k -NN graph Laplacian L (defined in Section 2.2), with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$.

The minimizers of the explicit MRIL objective (Section 2.2) and the implicit iMRIL objective (Section 2.3) have the following closed form expansions

$$f_{\text{MRIL}} = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j \quad \hat{f}_{\text{iMRIL}} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j$$

iMRIL's neighbor aggregation step applies the fixed spectral filter $\phi_{\text{iMRIL}}(\mu) = 1 - \mu$ to the graph Laplacian L , preserving low-frequency modes and suppressing high-frequency modes. The congruence between $\hat{f}_{\text{iMRIL}}$ and f_{MRIL} shows that iMRIL is equivalent to a built-in form of Laplacian smoothing (MRIL) with effective $\lambda \approx 1$ in normalized units. Unlike explicit regularization, this implicit filter requires no additional hyperparameter tuning.

Proof. From Lemma 3, the neighbor aggregation operator $S = D^{-1}A$ acts on Laplacian eigenmodes u_j as

$$Su_j = (1 - \mu_j)u_j,$$

where μ_j are the normalized Laplacian eigenvalues. Thus in the graph Fourier basis, neighbor aggregation corresponds to multiplying each mode by the fixed spectral filter $\phi_{\text{DARP}}(\mu) = 1 - \mu$.

On the other hand, Lemma 2 shows that explicit Laplacian regularization with parameter λ yields the spectral filter $\phi_\lambda(\mu) = (1 + \lambda \mu)^{-1}$. Both filters downweight high-frequency modes ($\mu \gg 0$) while preserving low-frequency modes ($\mu \approx 0$). The key difference is that $\phi_\lambda(\mu)$ requires tuning λ , whereas $\phi_{\text{DARP}}(\mu)$ is parameter-free.

To see the equivalence, note that for small μ ,

$$\phi_{\text{DARP}}(\mu) = 1 - \mu \approx (1 + \mu)^{-1} = \phi_{\lambda=1}(\mu) \quad \text{up to } O(\mu^2) \text{ terms.}$$

Thus DARP can be interpreted as performing Laplacian smoothing with an effective regularization weight of order $\lambda \approx 1$ in normalized units. Moreover, for large μ , $\phi_{\text{DARP}}(\mu)$ damps high-frequency modes even more strongly by driving them toward zero, providing a sharper low-pass effect than explicit regularization.

Therefore, DARP's aggregation step is mathematically equivalent to implicit Laplacian regularization with fixed spectral filter ϕ_{DARP} , eliminating the need to tune λ explicitly. $\square$

DARP may therefore be interpreted as a form of *locally adaptive implicit regularization*: in place of an explicit global weight λ , its neighbor-aggregation step enforces smoothness automatically through

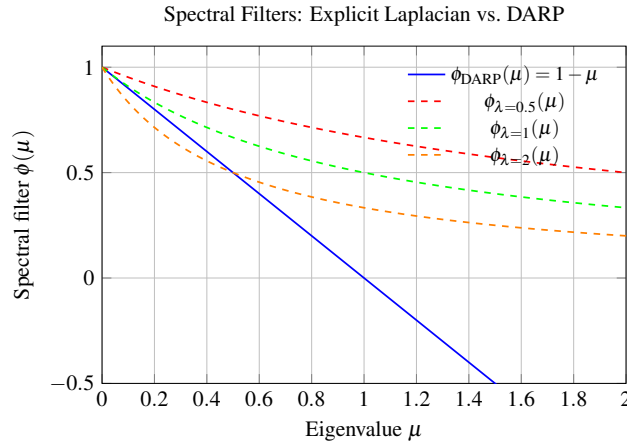


Figure 8: DARP attains a sharper low-pass spectral filter than explicit Laplacian regularization.

the graph structure. The effective regularization strength is a function of the local node degree and neighborhood geometry, and thus adapts to the local density of the expert demonstrations. Spectrally, this corresponds to the fixed filter $\phi_{\text{DARP}}(\mu) = 1 - \mu$, which attenuates high-frequency modes more aggressively than any fixed explicit λ . Figure 8 depicts this comparison, showing that DARP attains a sharper low-pass response without any hyperparameter tuning.

A.2 ADDITIONAL EXPERIMENTAL DETAILS

A.2.1 RETRIEVAL

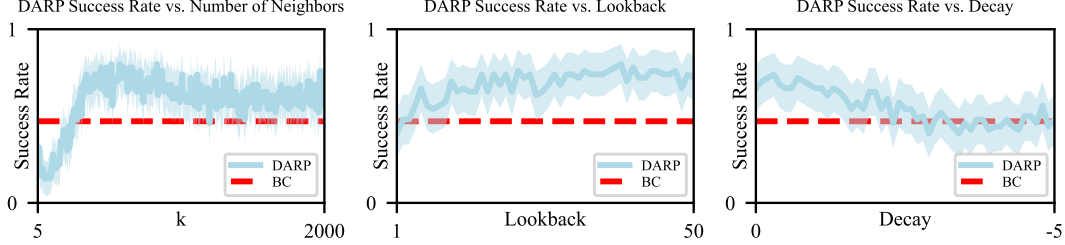


Figure 9: Sensitivity of DARP to the retrieval hyperparameters. (left) As a function of the neighbor count, performance is poor for small k , attains a global maximum near $k = 500$, and decays toward a plateau marginally above the BC success rate as k approaches the dataset size. (center) As a function of the history length, performance improves slightly as more past states are incorporated and falls below BC only when little or no history is used. (right) As a function of the exponential decay rate applied to older observations during retrieval, performance is sensitive to the decay: an excessively large decay effectively removes the lookback and reduces performance below BC. Success rate is measured over 50 trials on the Robosuite Stack environment, with 95% confidence intervals.

The choice of distance function $d(s_q, s_i^*)$ used to select the k neighbors is a critical design decision. While the plain Euclidean distance between states is sufficient in some cases, our experiments use a modified distance that exploits the sequential structure of the data by incorporating state history into the computation.

Let $S_q = (\dots, s_{q,-1}, s_{q,0})$ denote the query trajectory, with $s_{q,0}$ the current query state s_q . To evaluate $d(s_q, s_i^*)$ for an expert state s_i^* , we first identify the trajectory containing s_i^* , denoted S_j^* , and the index t of s_i^* within it, so that $s_i^* = s_{j,t}^*$. Given a lookback parameter ℓ specifying the number of past states to consider, we define

$$d(s_q, s_i^*) = \sum_{n=0}^{\ell-1} \|s_{q,-n} - s_{j,t-n}^*\|$$

i.e. the sum of Euclidean distances between the current and preceding $\ell - 1$ states of the query trajectory and the corresponding states of the source trajectory, over valid indices. Because more recent states should exert greater influence on neighbor selection, we introduce an exponential decay rate $\gamma \geq 0$ over the history:

$$d(s_q, s_i^*) = \sum_{n=0}^{\ell-1} \|s_{q,-n} - s_{j,t-n}^*\| \cdot e^{-\gamma n}$$

Figure 9 reports how the environment success rate varies as k , ℓ , and γ are swept.

A.2.2 CAN DARP HANDLE TASKS REQUIRING THE REPRESENTATION OF MULTI-MODAL ACTION DISTRIBUTIONS?

To evaluate DARP on complex action distributions, we test it on the Push-T task of (Chi et al., 2024), which requires representing multimodal action distributions (Figure 10). Here DARP uses a Set Transformer head that predicts the parameters of a Gaussian Mixture Model. The resulting GMM-headed DARP represents multimodal distributions effectively, yielding a 22% improvement over BC on Push-T (Q2) (Table 4). This confirms that DARP extends to multimodal action-distribution modeling requirements.

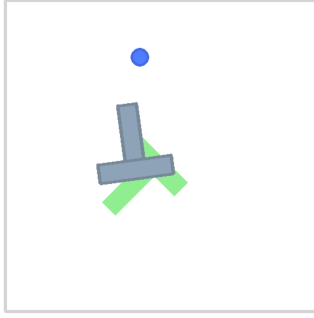


Figure 10: **Push-T Environment.** The agent controls the blue circle to push the T-shaped block to its target configuration.

Method	Score
BC	48 ± 8
DARP	70 ± 8

Table 4: **Push-T Results.** Mean score over 100 trials; DARP outperforms BC.

A.2.3 CAN DARP HANDLE DISCONTINUOUS ENVIRONMENTS WHERE NEARBY STATES MAY REQUIRE OPPOSING ACTIONS?

A principal concern for neighbor-based estimators is their behavior in environments with strong discontinuities, where states that are close in Euclidean distance may require markedly different actions. To stress-test DARP under this condition, we construct a modified version of the D4RL Umaze environment (Figure 11).

Even in this deliberately adversarial discontinuous environment, DARP attains a 57% success rate against BC’s 25% (Q3) (Table 5). This indicates that the difference-vector representation and permutation-invariant aggregation enable the model to discriminate between appropriate and inappropriate neighbors even when spatial proximity does not entail action similarity.

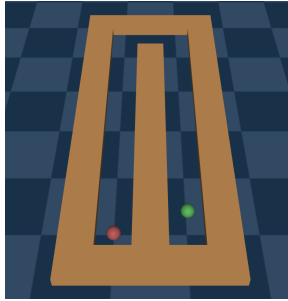


Figure 11: **Long maze environment.** The agent must drive a force-actuated ball from the green start location to the red goal location.

Method	Succ. (%)
BC	25
DARP	57

Table 5: **Long maze results.** Success rate over 100 trials; DARP substantially outperforms BC.

A.2.4 CAN DARP RECOVER FROM BC ERROR?

To analyze DARP’s robustness to accumulated error, we roll out a BC agent in an environment in which it is known to fail; every k steps we fork the environment and initialize a DARP agent from the forked state. The results (Fig. 12) show that, even as BC drifts away from the support of the expert demonstrations toward failure, DARP recovers and achieves high return. This corroborates the claim that DARP is robust to error accumulation and performs well on the marginally out-of-distribution states into which a failing BC agent drifts.

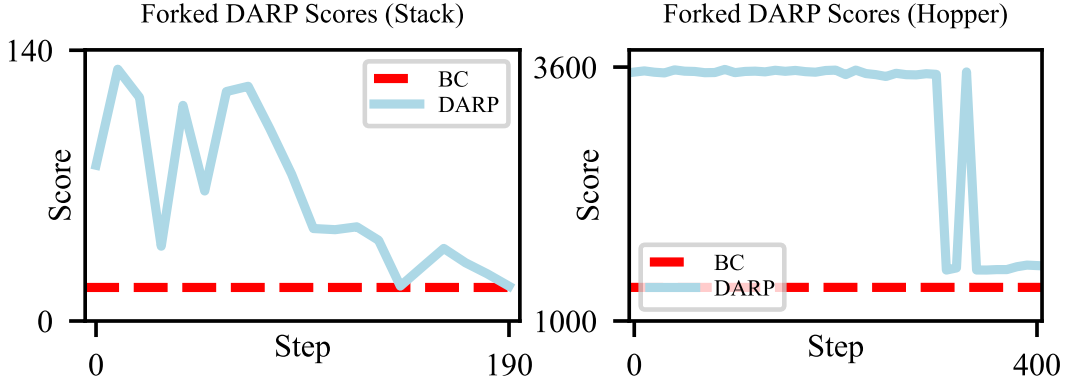


Figure 12: On two tasks (Robosuite Stack and MuJoCo Hopper), a BC agent is rolled out while the environment is forked every k steps ($k = 10$ here) to instantiate a DARP agent from the forked state. Even as BC approaches the end of its failing rollout, DARP continues to achieve high return, and only ceases to do so roughly halfway through the Stack rollout and roughly 80% of the way through the Hopper rollout.

Environment	Method	Improvement over BC (%)
Hopper	CCIL	32.6%
	DARP	53.2%
Walker	CCIL	84.1%
	DARP	84.1%
Ant	CCIL	12.6%
	DARP	84.5%
HalfCheetah	CCIL	5.4%
	DARP	418.7%

Table 6: **Relative improvement over BC: DARP versus CCIL.** Comparison on the standard MuJoCo benchmarks. DARP attains relative gains that are greater than or equal to those of CCIL on every task.

A.2.5 COMPARISON WITH CCIL

Table 6 compares DARP against CCIL, a baseline that induces smoothness explicitly. Using the scores reported in the CCIL paper, we compare the percentage improvement over BC. DARP outperforms CCIL by a substantial margin on three of the four environments, with the largest gap on HalfCheetah (418.7% versus 5.4% improvement).

A.2.6 DISTANCE METRIC SENSITIVITY

Method	Success Rate (%)
BC	47
DARP (Cosine Similarity)	70
DARP (Euclidean Distance)	75

Table 7: **Distance-metric sensitivity.** Success rates under R3M features. DARP performs comparably with Euclidean distance and cosine similarity, exceeding BC by 28% and 23% respectively. Results are on the Robosuite Stack task.

Although all experiments in this paper select neighbors via Euclidean distance, alternative metrics such as cosine similarity are natural candidates, particularly in high-dimensional embedding spaces

such as R3M. As Table 7 shows, substituting cosine similarity for Euclidean distance degrades DARP by only 5%, indicating that DARP is robust to the choice of distance metric.

A.2.7 DARP IN COMBINATION WITH DIFFUSION POLICY

Method	Stack	Threading	Peg Insertion
BC (MLP)	34	12	28
DARP (MLP)	62	38	40
Diffusion	52	44	54
DARP w/ Diffusion	82	72	66

Table 8: **DARP combined with a diffusion policy.** DARP yields substantial improvements over both the standard MLP and the diffusion policy backbones. Results are success rates over 50 trials on the Robosuite Stack, Threading, and Peg Insertion tasks under low-dimensional state. The demonstration count here is much smaller than in Table 2, so the BC and DARP entries are not directly comparable across tables.

Although all main-text experiments use an MLP backbone, diffusion policy (Chi et al., 2024) is a state-of-the-art policy class for imitation learning, particularly on manipulation tasks. Table 8 shows that DARP is complementary to diffusion and that the two can be composed into a more performant model. Instantiating DARP with a diffusion backbone improves success rate over the MLP-backbone DARP by 20%, 34%, and 26% on the three tasks respectively, exceeding standard BC by 48%, 60%, and 38%.

Method	Training (s/epoch)	Inference (s/step)
Diffusion	5.610	0.15700
DARP (MLP)	0.559	0.00437

Table 9: **Computational cost of DARP (MLP backbone) versus a diffusion policy.** The diffusion policy is markedly more expensive, incurring $\approx 10\times$ the per-epoch training cost and $\approx 36\times$ the per-step inference cost of DARP (MLP).

We further find empirically that DARP with an MLP backbone is substantially faster than the diffusion policy, most pronouncedly at inference time (Table 9).

A.2.8 DARP TRAINED ON HUMAN DEMONSTRATIONS

Method	Success Rate (%)
BC	45
DARP	60

Table 10: **Results under human demonstrations.** Success rates when training on human-collected data in place of RL-policy-collected data. Results are on the Robosuite Stack task.

The expert demonstrations used for the MuJoCo and Robosuite models are collected by a near-optimal reinforcement learning policy. It is important to verify that DARP retains its advantage over BC when trained on human-collected demonstrations, whose action distributions differ in character. On the Robosuite Stack task under human demonstrations, DARP outperforms standard BC by 15% (Table 10).

A.2.9 CHOICE OF RETRIEVAL HYPERPARAMETERS

Environment	Method	Score (Mean)
Hopper	BC	507.49
	DARP	805.69
Stack	BC	0.12
	DARP	0.31

Table 11: **Performance when validation loss selects the training epoch and retrieval hyperparameters.** Mean scores on the Hopper and Stack environments. DARP retains its advantage over BC under this selection protocol.

Figure 9 shows that certain retrieval-parameter settings (for instance, a very small neighbor count) cause DARP to underperform standard BC. Selecting the retrieval hyperparameters by minimizing validation loss is nonetheless an effective procedure for recovering performant settings (Table 11).

A.3 PSEUDOCODE

Algorithm 1 specifies the DARP training procedure in full.

Algorithm 1 Difference-Aware Retrieval Policies

```

1: Input: Expert demonstrations  $\mathcal{D}^* = \{(s_j^*, a_j^*)\}$ , number of neighbors  $k$ 
2: Initialize:  $f$  parameters  $\theta$ 
3: if  $g$  is parametric then
4:   Initialize:  $g$  parameters  $\psi$ 
5: end if
6: // Training Loop
7: while not converged do
8:   Sample batch of query data  $(s_q^*, a_q^*) \sim \mathcal{D}^*$ 
9:   for each query pair  $(s_q^*, a_q^*)$  in batch do
10:    // Find  $k$ -Nearest Neighbors from the entire dataset  $\mathcal{D}^*$ 
11:     $\mathcal{N}_k(s_q^*) \leftarrow \text{argmin-}k; d(s_q^*, s_j^*)$ 
12:    // Compute Neighbor-based Predictions
13:    for each neighbor index  $i \in \mathcal{N}_k(s_q^*)$  do
14:       $a'_i \leftarrow f_\theta(s_i^*, a_i^*, s_i^* - s_q^*)$ 
15:    end for
16:    // Aggregate Predictions
17:    if  $g$  is parametric then
18:       $\hat{a}_q \leftarrow g_\psi(\{a'_i\}_{i \in \mathcal{N}_k(s_q^*)})$ 
19:    else
20:       $\hat{a}_q \leftarrow g(\{a'_i\}_{i \in \mathcal{N}_k(s_q^*)})$ 
21:    end if
22:  end for
23:  // Update Parameters based on the batch loss
24:   $\mathcal{L} \leftarrow \sum_{(s_q^*, a_q^*) \in \text{batch}} \|\hat{a}_q - a_q^*\|^2$ 
25:  // Gradient descent step
26:   $\theta \leftarrow \theta - \alpha \nabla_\theta \mathcal{L}$ 
27:  if  $g$  is parametric then
28:     $\psi \leftarrow \psi - \alpha \nabla_\psi \mathcal{L}$ 
29:  end if
30: end while
31: Output: Trained parameters  $\theta$  and, if applicable,  $\psi$ 

```

A.4 RUNTIME ANALYSIS

Environment	k	Train (s/epoch)	Test (s/step)
Hopper (1,000 datapoints)	BC	0.102	0.00127
	100	0.126	0.00413
	200	0.128	0.00418
	300	0.130	0.00420
	400	0.131	0.00421
	500	0.130	0.00419
Stack (4,200 datapoints)	BC	0.431	0.00139
	100	0.556	0.00437
	250	0.539	0.00442
	500	0.559	0.00437
	750	0.576	0.00433
	1000	0.591	0.00478
	1500	0.682	0.00452
	2000	0.758	0.00451

Table 12: **Runtime across environments.** Per-epoch training time and per-step inference time (seconds) for Behavior Cloning (BC) and for varying neighbor counts k on the Hopper and Stack datasets.

Table 12 reports the computational overhead of DARP at both training and inference time. The measured cost scales sub-linearly in the neighbor count k , and inference time remains tractable for real-time robotic control. For instance, on the Robosuite Stack task at $k = 500$, per-step inference takes approximately 0.00437 seconds on our hardware, corresponding to a control frequency exceeding 230 Hz.